SE Mini Project

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BLOOD BANK MANAGEMENT SYSTEM

System Validation Report

Traceability to the supplied Software Test Plan v1.0

PASS	PASS	100	0.52 ms
Django system check	95 / 95 tests	Health smoke requests	Health smoke p95

Validation summary

The automated system validation covered core UI routes, the database-backed workflows, access controls, request and inventory state changes, alert/report pages, audit events, and the responsive/accessibility checks present in the repository. Django system checks pass and the migration state is consistent.

Plan requirement	Evidence	Status
BBMS-F-001 to F-004: registration, authentication, roles, profiles	Automated account, donor profile, login, inactive-user, RBAC and profile-edit tests	PASS
BBMS-F-005 to F-007: eligibility, slots, donation, inventory update	Booking eligibility and capacity tests; donation requires an in-progress appointment; unit is quarantined until cleared	PASS
BBMS-F-008: blood availability search	Search by blood group, component and location; only available, tested, unexpired units are returned	PASS
BBMS-F-009 to F-013: request, decision, reserve, issue and compatibility	Submission, approve/reject, exact group/component matching, atomic reserve/dispatch, expired/insufficient/incompatible cases	PASS
BBMS-F-014 to F-015: alerts, reports, audit and administration	Low-stock/expiry alert tests, real-data report tests, audit assertions and admin role tests	PASS
BBMS-NF-003: transaction and data integrity	Atomic reservation/issue tests and competing reservation TransactionTestCase	PASS
BBMS-SR-002 to SR-005: password, RBAC, session, audit	Password hash, inactive login, permission denial, session timeout/logout and audit tests	PASS
BBMS-SR-001: HTTPS/TLS	No TLS endpoint or deployment proxy is configured in this local environment	NOT RUN

Performance and scale

A local in-process Django test-client smoke made 100 requests to /health/: all returned HTTP 200; median 0.35 ms, p95 0.52 ms, maximum 47.85 ms. This endpoint-only, in-process measurement is not a network or representative user-workflow benchmark and does not establish BBMS-NF-001 for normal requests.

The transactional allocation test exercised two concurrent requests and passed. The BBMS-NF-004 target of 200 concurrent active users was not load-tested because the repository environment has no JMeter/load-test setup or production-like server profile. Monthly availability (BBMS-NF-002) cannot be established from this run.

Deployment security validation

manage.py check --deploy reported six development-configuration warnings: DEBUG enabled, development fallback SECRET_KEY, no HSTS duration, SSL redirect disabled, and session/CSRF secure-cookie flags disabled. These are expected for the local development configuration but must be resolved and revalidated for HTTPS production deployment.

Final status and limits

Django check: PASS. Migration drift check: no changes detected. Automated suite: 95/95 passed. No automated failure was recorded. Manual cross-browser/device UAT, external TLS verification, a 200-user load run, and monthly availability were not performed. The supplied plan is marked Draft and does not include detailed case scripts or a defect tracker, so formal UAT sign-off and the zero-open-critical-defect exit criterion cannot be independently certified.